DESIGN TASK

PROPOSE A ROBOTIC AUTOMATION COMMUNICATION FRAMEWORK

FOR INDUSTRIAL CONTROL

COURSE OUTCOME: CO3 – Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems

Bloom's Level: BL6 – Create / Design | Assessment Weightage: 35%

Academic Information	Details
Course Outcome	CO3
Bloom's Level	BL6 – Create
Assessment Type	Design / Model based on given requirements
Design Topic	Robotic Automation Communication Framework for Industrial Control
Primary Domain	5G/6G, Industrial IoT, Wireless Communication, Robotics
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Course	5g/6g Communication

Design Challenge

Industrial robots require fast, reliable and secure communication with PLCs, sensors, actuators, supervisory systems and other robots. The design challenge is to propose a communication framework that combines deterministic wired communication with flexible wireless connectivity while maintaining low latency, high reliability, synchronization, security and scalability.

Design Vision

The proposed framework follows an Industry 4.0 architecture in which time-critical control remains close to the machines at the edge, while monitoring, optimization and analytics can be provided through SCADA/HMI and cloud platforms.

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background

Modern manufacturing plants contain robotic arms, autonomous mobile robots, machine-vision systems, programmable logic controllers, sensors and intelligent actuators. These devices continuously exchange control commands and measurement data. Traditional isolated control networks are reliable but can become difficult to scale and integrate with wireless/mobile systems.

A robotic automation communication framework must therefore support two different communication classes. The first class is deterministic control traffic, where predictable delay and very low packet loss are essential. The second class is non-critical monitoring and analytics traffic, where throughput and scalability are more important than microsecond-level determinism.

1.2 Problem Statement

Design a hybrid communication framework capable of connecting multiple industrial robots, sensors, actuators, PLCs and supervisory systems. The framework should provide reliable real-time control, wireless mobility, secure data exchange, synchronization and a path toward 5G/6G-enabled smart manufacturing.

1.3 Design Objectives

- Achieve low and predictable communication latency for robot control.
- Provide reliable communication between robots, controllers, sensors and actuators.
- Support wired and wireless communication according to application requirements.
- Use spectrum-efficient wireless access for mobile and flexible robotic systems.
- Enable multi-robot coordination and resource sharing.
- Provide secure authentication, encryption and access control.
- Allow integration with SCADA, HMI, edge computing and cloud analytics.
- Provide a scalable architecture for future Industry 4.0 expansion.

1.4 Key Design Questions

1. Which communication technology should be used for time-critical robot control?
2. How can 5G/6G wireless access support mobile robots?
3. How should spectrum and radio resources be allocated among robots?
4. How can latency, reliability and synchronization be monitored?
5. Where should control decisions be processed: robot, PLC, edge or cloud?
6. How can the system remain operational during communication failures?

2. SYSTEM REQUIREMENTS AND DESIGN SPECIFICATIONS

2.1 Functional Requirements

- Robot controllers shall exchange command and status information with PLC/edge controllers.
- Sensors shall transmit position, force, temperature, proximity and machine-vision information.
- Actuators shall receive control commands from the appropriate controller.
- SCADA/HMI shall display machine status, alarms, production data and communication health.
- The edge layer shall perform local processing and protocol conversion.
- Wireless robots shall be dynamically admitted to the network.
- The system shall record communication events for diagnostics and maintenance.

2.2 Non-Functional Requirements

Parameter	Target Requirement	Reason
Control latency	< 10 ms	Fast robotic response
Critical packet loss	< 0.1%	Avoid incorrect control decisions
Availability	> 99.9%	Continuous factory operation
Synchronization	≤ 1 µs target	Coordinated motion and TSN/PTP
Wireless reliability	> 99.99% target for critical links	Industrial control continuity
Throughput	> 100 Mbps aggregate	Vision and monitoring traffic
Scalability	100+ devices	Large production cells
Security	TLS 1.3 / AES-256 where applicable	Protect industrial data
Recovery	Local fallback control	Maintain safe operation

2.3 Traffic Classification

Traffic Class	Examples	Priority	Preferred Link
Class A – Critical	Emergency/safety and closed-loop control	Highest	Industrial Ethernet / TSN
Class B – Real-time	Robot coordination, PLC status	High	Ethernet / 5G URLLC-style service
Class C – Monitoring	Sensor telemetry, diagnostics	Medium	Ethernet / Wi-Fi / 5G
Class D – Analytics	Logs, reports, cloud data	Low	MQTT / enterprise network

3. PROPOSED COMMUNICATION ARCHITECTURE

3.1 Layered Architecture

Layer	Devices / Technologies	Main Responsibility
Robot / Field Layer	Robots, sensors, actuators, I/O	Physical process interaction
Control Layer	PLC, robot controller	Real-time control logic
Edge / Gateway Layer	Industrial gateway, edge server	Protocol conversion, local analytics, buffering
Communication Layer	Ethernet, TSN, Wi-Fi 6, 5G	Data transport and resource management
Supervisory Layer	SCADA, HMI, MES	Monitoring, alarms and production supervision
Cloud / Enterprise	Cloud server, analytics	Long-term analytics, optimization and reporting

3.2 Proposed Architecture Flow

SENSORS / ROBOTS

↓

ROBOT CONTROLLER / PLC

↓

INDUSTRIAL ETHERNET / TSN ↔ Wi-Fi 6 / 5G

↓

EDGE GATEWAY / EDGE SERVER

↓

SCADA / HMI / MES

↓

CLOUD / ANALYTICS

3.3 Why a Hybrid Architecture?

A single communication technology cannot optimally satisfy every industrial traffic requirement. Industrial Ethernet and TSN provide deterministic behavior for fixed machines. Wi-Fi 6 and 5G provide mobility and flexible deployment. Edge computing keeps urgent decisions close to the robot, while cloud services are reserved for non-time-critical analytics.

3.4 Communication Interfaces

- Robot-to-controller: PROFINET, EtherCAT or equivalent industrial Ethernet.
- Controller-to-edge: Ethernet/TSN with OPC UA or industrial protocol gateway.
- Mobile robot-to-network: Wi-Fi 6 or private 5G.

4. WIRELESS COMMUNICATION AND SPECTRUM ACCESS DESIGN

4.1 Wireless Requirement

Mobile robots and reconfigurable production cells require wireless connectivity. The wireless design must support low latency, high reliability, mobility, interference management and efficient spectrum usage. A private 5G network is proposed for critical mobile-robot communication, with Wi-Fi 6 as an additional flexible connectivity option for suitable non-critical traffic.

4.2 Spectrum Access Strategy

Mechanism	Purpose	Application
Scheduled / configured resource allocation	Predictable access	Critical robot traffic
QoS-based priority	Protect high-priority packets	Robot control and alarms
Dynamic channel selection	Avoid interference	Wi-Fi/mobile robot links
Network slicing concept	Separate traffic classes	Control, monitoring, analytics
Admission control	Prevent network overload	Adding new robots
Beam management	Improve directional link quality	5G/mmWave future expansion

4.3 Proposed 5G Service Mapping

Industrial Traffic	5G Service Requirement	Design Treatment
Robot control	Very low latency + high reliability	Dedicated priority resources
Mobile robot coordination	Low latency + mobility	Private 5G QoS flow
Video / machine vision	High throughput	Separate high-bandwidth flow
Telemetry	Moderate throughput	Best-effort or standard QoS
Cloud analytics	Delay tolerant	Low-priority background traffic

4.4 Interference Management

- Use appropriate channel planning and frequency reuse.
- Monitor received signal strength and link quality.
- Move non-critical traffic away from congested resources.
- Use directional antennas/beamforming where appropriate.
- Maintain wired backup for critical fixed controllers.
- Set priority and QoS rules before adding new wireless devices.

5. PROTOCOL STACK AND DATA FLOW

5.1 Protocol Stack

Layer	Selected Technology	Function
Application	OPC UA / MQTT	Industrial data and telemetry
Transport	TCP / UDP	Reliable or low-overhead transport
Network	IPv4 / IPv6	Logical addressing and routing
Link / Industrial	Ethernet / TSN / Wi-Fi 6 / 5G	Data transmission
Synchronization	IEEE 1588 PTP	Precise device timing
Security	TLS 1.3, certificates, RBAC	Authentication and confidentiality

5.2 Data Flow – Control Path

1. Sensor measures position, force or process condition.
2. Sensor data reaches the robot controller or PLC.
3. The controller executes the control algorithm.
4. Control commands are transmitted to the actuator/robot.
5. Robot motion changes the physical process.
6. Status feedback is returned to the controller.
7. The edge layer monitors the control loop without becoming a single point of failure.

5.3 Monitoring Path

Non-critical measurements are copied or published from the edge layer to SCADA/HMI. Selected information is then forwarded to an MQTT broker or cloud platform for historical analysis, predictive maintenance and production optimization.

5.4 Failure Handling

Failure	Detection	Response
Wireless link degradation	RSSI/quality monitoring	Switch resource/channel or use backup
Edge connection loss	Heartbeat timeout	Continue local control
Cloud unavailable	Application timeout	Store data locally and retry
Controller communication failure	Watchdog	Enter predefined safe state

6. HARDWARE AND SOFTWARE DESIGN

6.1 Hardware Components

Component	Suggested Specification / Example	Quantity
Industrial robot	6-axis robotic arm or educational robot	2+
Robot controller	Industrial/embedded controller	2+
PLC / edge controller	Industrial PLC or IPC	1–2
Industrial gateway	Dual Ethernet + Wi-Fi/5G capability	1
Sensors	Proximity, force, temperature, vision	Multiple
Actuators	Servo motors, gripper, valves	Multiple
Industrial switch	Managed Ethernet/TSN-capable switch	2+
Wireless access	Wi-Fi 6 AP / private 5G infrastructure	1+
SCADA/HMI	Industrial PC or visualization panel	1
Edge computer	Industrial PC / embedded Linux system	1
Time source	PTP-capable clock / grandmaster	1

6.2 Software Components

- PLC programming environment for control logic.
- Robot programming environment for motion and task control.
- OPC UA server/client for standardized industrial data exchange.
- MQTT broker for telemetry and event distribution.
- SCADA/HMI software for visualization and alarms.
- Edge analytics software for local processing.
- Network monitoring tools for latency, packet loss and utilization.
- Simulation tools such as MATLAB/Simulink, NS-3 or Packet Tracer where applicable.

7. CONTROL, RELIABILITY AND SECURITY DESIGN

7.1 Real-Time Control Strategy

The proposed system uses a hierarchical control strategy. The robot controller and PLC perform the fastest control functions locally. The edge layer performs coordination and optimization. SCADA/HMI provides supervisory control, while cloud services are used for long-term analytics. This prevents cloud delay from affecting the primary safety and motion-control loop.

7.2 Reliability Techniques

- Redundant communication paths for important infrastructure.
- Heartbeat messages between critical devices.
- Watchdog timers for controller communication.
- Priority queues and QoS classification.
- Local fallback operation when cloud connectivity fails.
- Buffering and retransmission for non-critical telemetry.
- Continuous monitoring of packet loss and latency.

7.3 Security Architecture

Security Layer	Technique	Purpose
Device identity	Certificates / unique credentials	Authenticate devices
Communication	TLS 1.3 where applicable	Protect data in transit
Network	Segmentation / VLAN / firewall	Isolate critical systems
Access	Role-Based Access Control	Limit operator privileges
Application	Secure API / authenticated MQTT	Protect data services
Monitoring	Logs / alerts	Detect suspicious behavior
Maintenance	Signed updates	Prevent unauthorized software

7.4 Safety Consideration

Communication security and functional safety are complementary. Emergency-stop functions should use an appropriate safety-rated mechanism rather than depending only on a general-purpose wireless network. The communication framework should support safe-state transitions whenever a critical control link is lost.

8. PERFORMANCE ANALYSIS AND EVALUATION PLAN

8.1 Performance Metrics

Metric	Measurement Method	Target
End-to-end latency	Timestamp command and feedback	< 10 ms control target
Jitter	Calculate variation of packet delay	Low and bounded
Packet loss	Lost packets / transmitted packets	< 0.1% critical traffic
Throughput	Traffic volume / time	> 100 Mbps aggregate target
Availability	Operational time / total time	> 99.9%
Synchronization error	PTP clock difference	≤ 1 µs target
Wireless reliability	Successful packets / total	> 99.99% target for critical link
Recovery time	Failure to restored service	As low as practical

8.2 Experimental Scenarios

1. Normal operation with two robots and one PLC.
2. Multiple robots simultaneously transmitting status information.
3. Wireless robot moving between access coverage zones.
4. Network congestion caused by high-volume machine-vision traffic.
5. Temporary wireless interference.
6. Edge-to-cloud connection failure.
7. Controller/network device failure and recovery.
8. Addition of new robots to evaluate scalability.

8.3 Proposed Simulation / Prototype Procedure

- Create the network topology using an appropriate simulation or laboratory platform.
- Assign separate traffic classes for control, monitoring and analytics.
- Generate periodic robot-control and sensor packets.
- Measure latency, jitter, throughput and packet loss.
- Increase the number of robots and observe network utilization.
- Introduce interference or congestion and evaluate QoS behavior.
- Compare wired-only, wireless-only and hybrid architectures.

9. IMPLEMENTATION PLAN AND COST-LEVEL DESIGN

9.1 Implementation Phases

Phase	Activities	Output
1. Requirement study	Define robots, traffic and performance targets	Requirement specification
2. Architecture design	Select topology and communication technologies	Network architecture
3. Hardware setup	Connect robots, PLC, sensors and gateway	Physical testbed
4. Network configuration	IP, VLAN, QoS, TSN/5G/Wi-Fi settings	Operational network
5. Protocol integration	OPC UA/MQTT and controller interfaces	Interoperable system
6. Monitoring	SCADA/HMI and network dashboards	Visualization
7. Testing	Latency, reliability and security tests	Performance report
8. Optimization	Tune QoS and resource allocation	Final design

9.2 Indicative Cost Categories

Category	Examples	Cost Level
Robotic equipment	Robot arm, controller, gripper	High
Industrial control	PLC, I/O modules	Medium–High
Networking	Managed switch, gateway	Medium
Wireless	Wi-Fi 6 / private 5G equipment	Medium–High
Sensors	Proximity, force, vision	Low–High
Computing	Edge PC, HMI/SCADA PC	Medium
Software	SCADA, simulation and analytics	Low–High depending on license

9.3 Low-Cost Academic Prototype

For an academic prototype, the full industrial robot and private 5G infrastructure can be represented by a microcontroller/robotic arm, PLC or industrial controller, Raspberry Pi/industrial PC, Wi-Fi 6 router, managed switch and simulated 5G traffic. The same architecture and performance metrics can be demonstrated without reproducing a complete factory.

10. DESIGN INNOVATION AND CO3 MAPPING

10.1 Creative Design Features

- Hybrid wired + wireless architecture instead of relying on one communication medium.
- Priority-aware spectrum/resource allocation for robotic traffic.
- Edge-first control architecture to minimize dependence on cloud connectivity.
- Traffic isolation between safety/control, monitoring and analytics.
- Dynamic admission control for adding new robots.
- Predictive network-health monitoring using edge analytics.
- Future-ready integration with 5G/6G technologies such as advanced beamforming and intelligent resource management.

10.2 Mapping to CO3

CO3 Requirement	How the Proposed Design Addresses It
Design spectrum access strategies	5G/Wi-Fi resource allocation, QoS, channel selection and admission control
Waveform / wireless transmission awareness	OFDMA-based time-frequency resource allocation and 5G physical-layer concepts
Multiple access techniques	Multiple robots share radio resources using scheduled/OFDMA-based access
Efficient wireless transmission	Priority, QoS, interference management and adaptive resource assignment
Application of design skills	Complete industrial communication architecture with measurable targets

10.4 Expected Academic Learning

- Understanding industrial communication requirements.
- Applying multiple-access and spectrum-management concepts.
- Designing communication layers for real-time systems.
- Evaluating latency, reliability and throughput.
- Connecting 5G/6G concepts to a practical industrial application.
- Developing engineering trade-off and system-design skills.

11. EXPECTED RESULTS, APPLICATIONS AND CONCLUSION

11.1 Expected Results

- A scalable communication architecture for connected industrial robots.
- Low-latency control for fixed robotic cells.
- Flexible wireless connectivity for mobile robotic systems.
- Efficient sharing of radio resources among multiple devices.
- Improved network reliability through QoS and redundancy.
- Secure exchange of industrial control and monitoring information.
- Continuous SCADA/HMI visibility and edge-based diagnostics.
- A practical path toward 5G/6G-enabled smart manufacturing.

11.2 Industrial Applications

Application	Communication Requirement	Framework Benefit
Automotive assembly	Deterministic robot coordination	Low-latency control
Warehouse robots	Mobility and handover	5G/Wi-Fi connectivity
Machine vision	High throughput	Dedicated high-bandwidth traffic
Packaging	Fast repetitive control	Priority scheduling
Welding robots	Precise synchronization	TSN/PTP + controller
Predictive maintenance	Continuous telemetry	MQTT + edge/cloud analytics
Smart factory	Large device population	Scalable architecture

11.3 Final Design Statement

The proposed robotic automation communication framework combines industrial Ethernet, TSN, Wi-Fi 6, private 5G, edge computing, OPC UA and MQTT into a layered architecture for industrial control. Time-critical decisions are performed locally, while wireless and enterprise connectivity provide mobility, monitoring and analytics. Spectrum-aware resource allocation, QoS, synchronization, security and failure handling make the framework suitable for a modern smart factory.

11.4 Conclusion

The design satisfies the BL6 requirement by creating a complete communication solution rather than simply describing existing technologies. It identifies requirements, selects technologies, proposes a layered architecture, defines spectrum and multiple-access strategies, establishes measurable performance targets and provides an evaluation plan. The resulting framework demonstrates how CO3 concepts can be applied to a real industrial automation problem.